

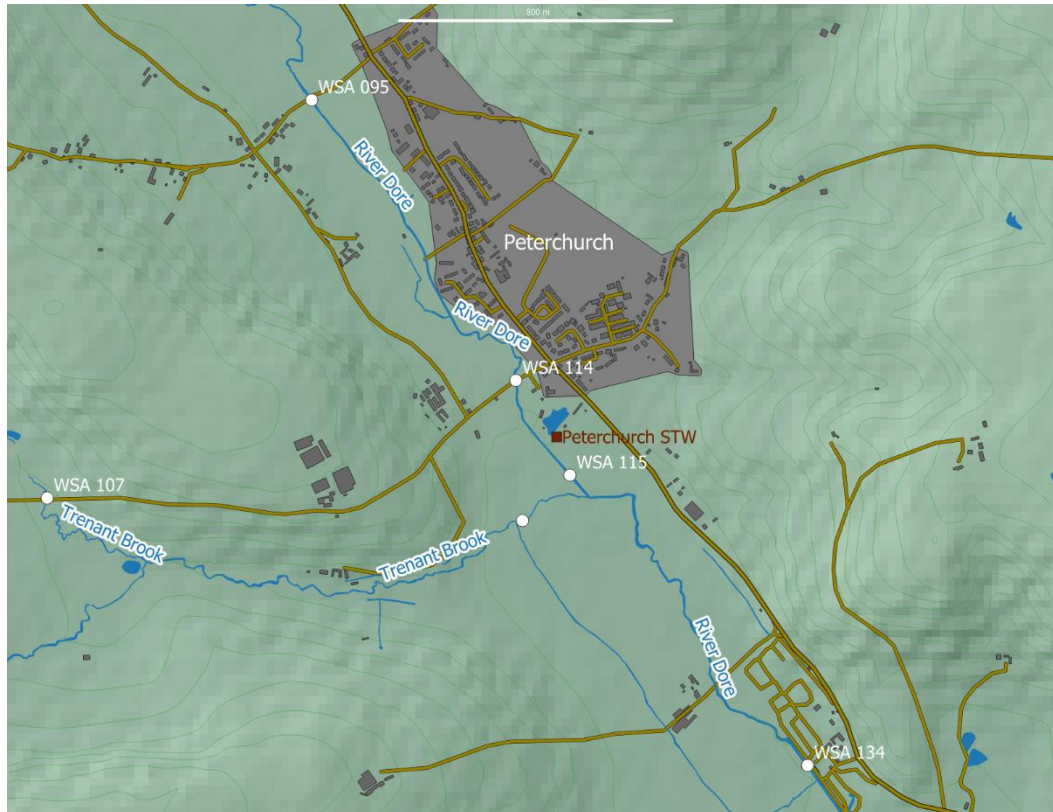


Hot-spot report: River Dore at Peterchurch

A report prepared on behalf of Wye Salmon Association (WSA)

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Location



Map of Peterchurch and the adjacent River Dore. White circles are WSA sampling sites

Peterchurch is situated adjacent to the River Dore. The village is served by Peterchurch STW (operated by DC-WW; input load 856 PE), discharging into the Dore at the downstream end of the village.

Hypothesis

That Peterchurch STW exports unacceptable levels of phosphorus and other pollutants into the River Dore.

Test

WSA citizen scientists have monitored reactive phosphorus (RP) and electrical conductivity (EC) at four sites around Peterchurch. Fairfield Bridge (WSA 114) lies immediately above the STW. In March 2023, WSA 115 was discontinued in favour of Poston Mill (WSA 134). Therefore, the data from WSA 114 and WSA 134 will be compared. Paired samples from these sites have been measured within 30 minutes of each other, approximately weekly, over the period Apr-23 to Aug-23.

Results

The results from 18 paired samples are summarised in the table below:

Parameter	Upstream (WSA 114)	Downstream WSA 134)	t-test
Mean RP (mgP/l)	0.04	0.09	0.00%
Mean EC @25°C (µS/cm)	477	513	0.10%

Mean values for reactive phosphorus and electrical conductivity on the River Dore at locations upstream and downstream of Peterchurch STW. The 2-sided, paired t-test gives the probability that the two mean values belong to the same population (i.e., no change).

This tells us that there is a huge (x2) increase in mean RP between the two sites, which is statistically significant with over 99.9% confidence. Mean EC increases by 7%, also with 99.9% confidence.

Of relevance is the confluence of Trenant Brook between the STW and WSA 134. The 2023 mean RP at WSA 107 on Trenant Brook was 0.03 mgP/l. Although the downstream site closer to the confluence has been discontinued for 2023, we know that the 2022 mean RP concentration there was 0.04 mgP/l. So, the Trenant Brook is relatively clean and will be serving to reduce RP levels at Poston Mill. Thus, the apparent increase of 0.05 mgP/l due to the STW is undoubtedly an underestimate of the situation immediately below the STW. Indeed, the 2022 mean RP measured at WSA 115 was 0.40 mgP/l and included five samples where the Hanna maxed out at 0.82 mgP/l. According to Rivers Trust, the SSO at Peterchurch STW spilled 12 times in 2022, which may go some way to explaining the high readings.

Conclusion

- The 2x increase in River Dore RP concentration resulting from the discharge from Peterchurch STW is large and has extremely high statistical confidence. Upstream of the STW, the River Dore's RP concentration meets the requirements for WFD *good* classification (<0.064 mgP/l) over the period monitored. Below the STW, the RP concentration corresponds to a *moderate* classification only. Thus, the STW is causing non-compliance in this section of river, which is a cause for concern. We are aware of no plans to introduce P-stripping technology, or any other improvements at this site over the next few years (AMP7 & AMP8).
- The increase in electrical conductivity is equally alarming. Given the fact that most of the EC signal results from natural mineral ions, a 7% increase represents a very significant pollution load. We have separately estimated that the discharge from Peterchurch STW in normal operation represents approximately 4% of Q95 flow. This is a very significant volume of effluent.